Semantic Segmentation With Attention Mechanism for Remote Sensing Images

Qi Zhao, *Member, IEEE*, Jiahui Liu, Yuewen Li, and Hong Zhang

***Abstract*— Semantic segmentation for high-resolution remote sensing images is one of the most significant tasks in the field of remote sensing applications. Remote sensing images contain substantial detailed information of ground objects, such as shape, location, and texture. Therefore, these objects make the images exhibit large intraclass variance and small interclass variance, which makes it very difficult to be recognized. In this study, an end-to-end attention-based semantic segmentation network (SSAtNet) is proposed. A pyramid attention pooling module is proposed to introduce the attention mechanism into the multiscale module for adaptive features refinement. To correct the detailed information, the pooling index correction module integrates pooling index maps from the encoder with high-level feature maps, which can help recover the fine-grained features. In the encoder phase, a more effective ResNet-101 backbone is designed to capture detailed features. What is more, a series of data augmentation methods are proposed to enhance the model's robustness. The proposed model is compared with several previous advanced networks and achieves the state of the art on the ISPRS Vaihingen dataset. The experiment results prove the effectiveness of the SSAtNet.**

***Index Terms*— Attention mechanism, convolutional neural networks (CNNs), multiscale module, remote sensing, semantic segmentation.**

I. Introduction

SEMANTIC segmentation of high-resolution aerial imagery is one of the fundamental tasks of remote sensing applications. It is also one of the most challenging tasks in the field of remote sensing. Since it is not only extracting object boundaries, it labels every pixel in an image to a specified category. For the urban scenes, the problem becomes more complicated because of the similarity among inhomogeneous objects and the diversity of homogeneous objects. Thus, semantic segmentation technique plays a significant role in our society, such as self-driving, agricultural statistics, and even military purposes.

The traditional methods for semantic segmentation used the low-level features extracted from color, shape, and texture. Then, these features would be clustered and classified in high-level space for the whole image segmentation. It could be seen the disadvantages of such a method: 1) the model performance depends heavily on the artificial designed features, which makes the model expression limited and 2) these low-level features are not expressive enough for the image segmentation tasks. Thus, the abstract features or even the hierarchical features need to be extracted from the image for the vision tasks.

In recent years, a convolutional neural network (CNN) has been proven effective in lots of computer vision tasks, such as classification, detection, and segmentation [1], [2]. The CNN method extracts the hierarchical features of different objects in an image automatically. It can capture the detailed features in the shallow layers of the network and complicated features in the deeper layers. For image segmentation, a fully convolutional network (FCN) [3] is proposed to do the dense prediction of an image, which predicts the label of each pixel of the image. Since the FCN boosts the segmentation accuracy greatly, it now becomes the normal framework for image semantic labeling tasks.

Besides, lots of FCN-based models are proposed to improve the model performance on the standard dataset, such as U-Net [4] and SegNet [5]. To capture both the tiny and huge object features of the image, multiscale techniques are combined in these models. U-Net adopts skip connections from shallow layer to deep layers; SegNet uses an encoder–decoder framework with up-pooling skills to improve the accuracy. However, due to the subsampling in CNN models, the final feature maps resolution is much coarser than the input image, losing lots of detailed information. Except for the model framework, some researchers paid attention to the loss function. Focal loss [6] and OHEM [7] are proposed for mining small and hard samples. Milletari *et al.* [8] proposed multinomial loss function to optimize the Jaccard index directly.

The remote sensing image, with high-intraclass variance and low-interclass variance, is much difficult to be labeled than standard dataset images. To solve the problem, contextual information will be captured to cope with the complicated scene. Thus, the multiscale module design is more significant in this article. There are four popular multiscale methods: image pyramid [9], [10], encoder–decoder framework [4], [5], [11]–[15], using context module [16]–[20], and spatial pyramid pooling (SPP) [21]–[23]. In this article, the latter three methods are adopted to build the entire model. The image pyramid methods are ignored in this article because the image pyramid method is much more space and time-consuming during the training phase.

Furthermore, attention mechanisms have been successfully applied to semantic segmentation [24]. The attention module

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Qi Zhao, Jiahui Liu, and Yuewen Li are with the School of Electronics and Information Engineering, Beihang University, Beijing 100191, China.

Hong Zhang is with the School of Astronautics, Beihang University, Beijing 100191, China (e-mail: dmrzhang@buaa.edu.cn).

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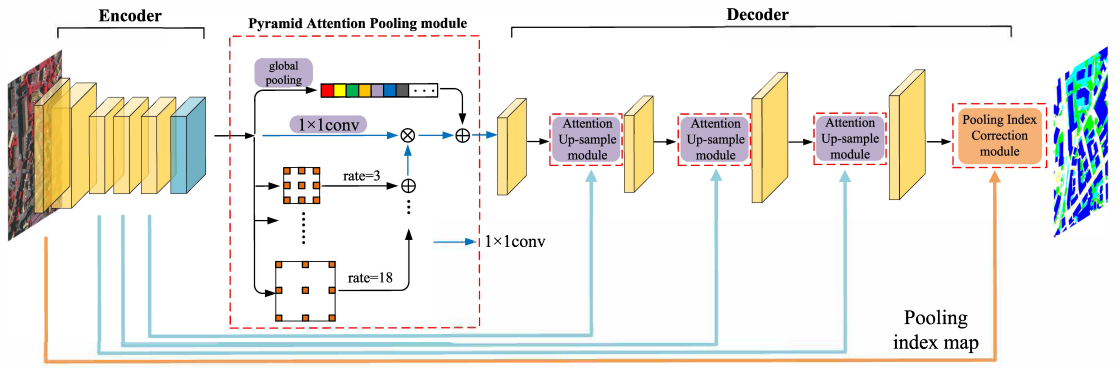


Fig. 1. Overview of the proposed SSAtNet.

can introduce the model to focus more on the meaningful features, both channelwise and spatialwise. In our model, the attention mechanism is combined with a multiscale module to improve the accuracy of labeling high-resolution aerial imagery. A new spatial attention pyramid pooling module is proposed in this article.

Another motivation for our work is that lots of papers do not pay much attention to data augmentation. Deep learning models depend heavily on training data. However, the dataset that we use, ISPRS [25], does not has many training images. Thus, the main challenge of the remote sensing task is the lack of training image data. The data augmentation method is one of the contributions of this article.

Thus, the main contributions are concluded as follows.

1) The attention mechanism is applied to the general ASPP module to improve model performance. Rather than concatenation, the proposed module uses the output from the convolutional layer of a large dilation rate to guide the small one. Besides, the channelwise attention module could also help the model to capture the hierarchical information.
2) A more effective backbone is proposed to capture the detailed information. Some layers in the raw ResNet are adjusted because of the information divulged in the original structure.
3) A pooling index correction (PIC) scheme is proposed, which adds the up-sampled feature maps with the pooling index map from the encoder, to restore the fine details.
4) A series of data augmentation methods are proposed, including random sampling, random rotating, and random scale.

The rest of this article will be organized as follows. Section II introduces the related works. The model detail will be explained in Section III. The experiment setting and the result will be shown in Section IV. To gain more insight into the proposed model, the ablation analysis will be discussed in Sections V and VI. Finally, in Section VII, we will give the conclusion of this article.

II. Related Work

After AlexNet [1] improved the performance of the computer vision task greatly in the ILSVRC 2012, CNN has become the mainstream method in the computer vision field. Benefit from the progress of computer hardware compute ability, more complex and bigger models are proposed to obtain higher accuracy in image classification fields in recent years. VGGNet [26] used multiple convolutions to improve the model performance; GoogleNet [27] introduce the inception module, which expands the model width, and use an average pool instead of the fully connection layer; and ResNet [28] adopted shortcut connections to solve the gradient vanishing problem. The ResNet-101 structure is adopted as the backbone in our paper. DenseNet [29] and CondenseNet [30] made full use of feature reuse to surpass state of the art in 2018. These networks were usually adopted as the backbone in models for complex vision tasks, such as image recognition and segmentation.

The fully connection network [3] proposed a brand new framework for semantic segmentation task, which replaced the fully connected layers with 1×1 convolution in VggNet. Thus, the model could do the dense prediction tasks. However, the output feature maps are extremely coarse, which means lots of information are lost. Besides, FCN adopted a single up-sample layer to recover the output score map, which made the final score map worse. Low-level details information that is missing makes FCN performance limited.

The first tricky problem is focused on how to produce more accurate dense predictions from the coarse feature maps. Because the final output feature maps contain only abstract information, the low-level details are essential for the model performance. Meanwhile, remote sensing images are more sensitive to the object boundary, shape, and corner. To solve the problem of low-level information missing, U-Net used skip connections to combine the deeper feature maps with shallow feature maps, which reused low-level feature maps to help recover more information. SegNet proposed an encoder–decoder framework and used up-pooling with recorded pooling index to recover the distinct information, such as shape and edge. Meanwhile, SegNet introduced a large number of parameters and computations. Multiple models adopted deconvolution [31], up-sampling, and up-pooling skills in the decoder phase. These models fuse the information from shallow and deep layers to refine the output accuracy. However, they pay no attention to the feature information both in a different layer and a different location. Since features at different locations and depths have different importance, the feature maps in different

channels and different locations could not be done the adding operation directly.

Another tough problem is the recognizing incoherence of the same but confusing objects for the remote sensing images of high-intraclass variance and low-interclass variance. To solve the problem, multiscale and attention techniques are developed. The multiscale method can be divided into four categories, including image pyramid, encoder–decoder framework, context module, and SPP. Image pyramid methods input multiscaled images into the same model with shard weight and bias. The output feature map of different scaled images could capture a different range of contextual information. Finally, the feature maps of multiscaled input will be fused. Such a method consumes lots of GPU memory and training time. This method is abandoned in this article. The encoder–decoder-based model contains two parts. The encoder captures hierarchical features from shallow layers to deep layers. Meanwhile, the longer range contextual information can be captured gradually by the deeper layers. Because of the down-sampling layer in the encoder, the output feature map will be coarser than the input image. The decoder recovers the low-resolution feature map to the dense prediction as accurate as possible. The main problem of this method is resolution loss. The encoder with dilated convolution solved the problem perfectly. Compared with the standard convolution, dilated convolution has a larger receptive field, which means that it can combine long-range pixel context. Meanwhile, the size of output feature maps can maintain stability, which means that the resolution is much higher than the standard encoder structure. ResNet-101 with a dilated convolution structure is adopted in this article. The image-level features are exploited in ParseNet [32] for global context information. DeepLabv2 [22] proposes atrous SPP (ASPP), where parallel dilated convolution layers with different rates capture multiscale information. Recently, the pyramid scene parsing model [23] performs spatial pooling at several grid scales and demonstrates outstanding performance on several semantic segmentation benchmarks.

For attention mechanism, Chen *et al.* [24] proposed the channelwise attention module and the spatialwise attention module to combine features with different weights, which could help the model focus on the appropriate channel and certain area on a single channel. Besides, PANNet [33] introduced the attention module in the multiscale module: introducing a feature pyramid attention module to perform a spatial pyramid attention structure on high-level output and combine global pooling to learn a better feature representation. Besides, a global attention up-sample (AU) module on each decoder layer to provide global context as guidance of low-level features to select category localization details.

Many groups also focus on the semantic segmentation tasks of remote sensing images [34]–[37]. Pan *et al.* [38] adopted DenseNet as a model backbone and used SPP as a multiscale module. To make the model pay more attention to small and difficult categories, focal loss [6] is used to replace the standard cross-entropy loss. ReSegNet [39] proposed ensemble training and inference strategies to suppress the adverse consequences of the encoder–decoder framework, which is state of the art. There are some works that pay more attention to the use of contextual information. Nogueira *et al.* [40] train a dilated network with distinct patch sizes to capture multicontext characteristics from heterogeneous contexts. Mou *et al.* [41] introduce the spatial relation module and the channel relation module to learn the global relationships between two spatial positions or feature maps, which can produce relation-augmented feature representations. Zhou *et al.* [42] add a feature decoupling module, following the CNNs, to encode the class co-occurrence relations into the convolutional features.

In this work, the attention-based multiscale module is proposed to improve the model performance in remote sensing images. More specifically, the pyramid attention pooling (PAP) module is proposed for multiscale context capturing; the AU module is adopted to help high-level features enhance the contextual information from low-level features from the encoder, and the PIC module is proposed to recover the information losing in pooling layer. Besides, the ResNet backbone is refined to make our model more expressive.

III. Methodology

In this section, the proposed semantic segmentation network (SSAtNet) will be introduced. Similar to the analysis in Section I, the high-level features, which are captured by the CNN model, are good at classifying but not appropriate for the dense prediction tasks, such as semantic segmentation. Based on the encoder–decoder network framework, the proposed modules help to refine the multiscale context.

In the encoder phase, a more effective ResNet-101 backbone is proposed to capture the hierarchical information, from shallow to deep. The output high-level features will be fed into the PAP module, which integrates the multiscale context information, to refine the quality of feature maps. In the decoder phase, the proposed AU module will be used to recover as more accurate information as possible from the shallow layer in the encoder. Besides, the PIC module will use a pooling index map, which is reserved in the encoder phase, to describe the details more precisely.

The overview of the proposed model is shown in Fig. 1.

In this section, the proposed effective ResNet-101 backbone, the PAP module, the PIC module, and the attention-based up-sampling module will be introduced orderly.

A. ResNet Refinement

ResNet is proposed for image classification tasks. Since the classification task is easier than the segmentation tasks, lots of detailed information on the feature can be abandoned. However, this is harmful to the segmentation purpose. Thus, the raw ResNet-101 architecture is improved in this work:

1) The first 7×7 convolution layer is replaced by a sequence of three 3×3 convolution layers.
2) The standard convolutions are replaced by dilated convolutions.
3) Adding a global pooling layer in the second stage of ResNet-101 structure.

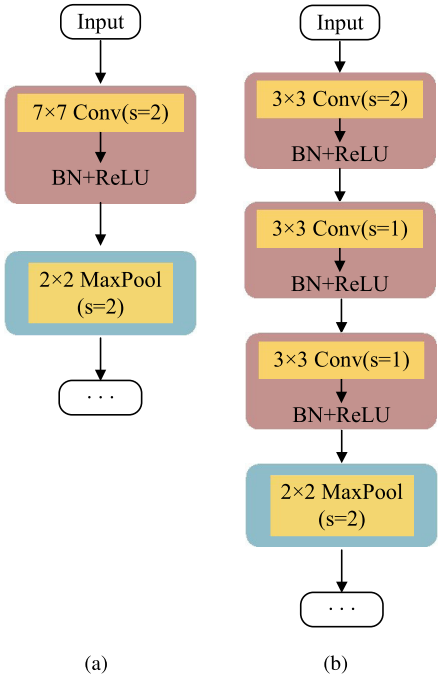


Fig. 2. Residual block refinement. (a) Standard ResNet structure. (b) Replace the first 7×7 convolution layer with three 3×3 convolution layers.

1) Small Convolutional Kernel: A classic ResNet network consists of input 7×7 convolution layer and four following stages, which contains residual blocks with different layers. However, the input convolutional layer has a 7×7 filter, with the stride of 2 and 64 output channels. The following layer is a 2×2 max-pooling layer with a stride of 2. Empirically, the 7×7 convolution layer blurs the fine details of the raw image and brings many computations. Similar to SENet [43] and PSPNet, we also replace the first 7×7 convolution layer with a 3×3 convolution layer, with a stride of 2. Two 3×3 convolutions with a stride of 1 are added. Such a structure can capture more detailed features and save computations. Fig. 2 shows the above adjustment.

2) Encoder With Atrous Convolutions: The raw ResNet network does the down-sampling by 32 times from input images to the output feature maps. However, a large number of pixels are dropped, which makes it more difficult to recover details in the decoder phase. Thus, similar to Deeplab v3 network, dilated convolutions [15] are introduced into the backbone, with an eight times down-sampling rate from the raw image to the feature maps, i.e., the output stride is 8. The dilation rate is set to [1, 2, 2, 4] in the different stages of the ResNet structure. It could be inferred that the feature map size will be maintained in stages 2, 3, and 4. For two-dimensional feature maps, the pixel location i in output feature map y can be performed by dilated convolution on input feature maps x, and kernel parameters w can be presented as

$$y(i) = \sum_{i=k} x(i + r \times k) \times w(k) \tag{1}$$

where r is the dilation rate, corresponding to the sampling stride in input feature maps. Fig. 3 presents the diagram of dilated convolution.

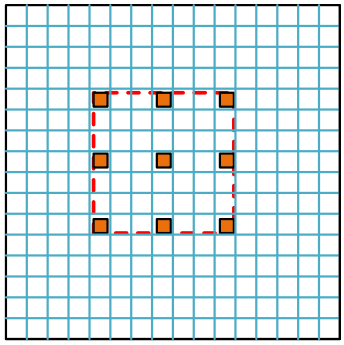


Fig. 3. Encoder with atrous convolution.

3) Adding a Global Pooling Layer: At the beginning of the first residual block in each stage, there is a down-sampling module, which replaces the shortcut connection. In our model, because the dilated convolutions are adopted, there will be no down-sampling module in stages 1, 3, and 4. Only stage 2 contains the down-sampling module, which has a 1×1 convolution with stride 2. In the down-sampling block, there are two computing paths. One path (A) has three convolutions, whose kernel sizes are 1×1, 3×3, and 1×1. The first 1×1 convolution has a stride of 2, and the number of output feature map channels will be four times the input feature map channel number. Another path (B) converts the input shape to be the shape of the output of path A by a 1×1 convolution with the stride of 2. However, this means that half of the pixels on the former feature maps will be ignored. The pixels on the output feature maps should have a bigger receptive field. To solve the problem and not increase the model computation, an average pooling layer in front of the 1×1 convolution was added. Fig. 4 shows the change.

B. PAP Module

The original ASPP model proposed in [22] has four parallel convolutions with different dilation rates, which are applied to the final feature maps of the encoder. Such a parallel scheme has been proven successful in practice. By the way, the dilated convolutions in the ASPP module may result in local detailed information losing and "grid phenomenon," which is harmful to the quality of output feature maps. The SPP module in PSPNet is also in such a scheme, which motivates the idea of ASPP. However, PSPNet drops too many pixels, which means that much information that needs to be restored in the decoder phase is lost. Thus, in our proposed model, we choose the ASPP module as the baseline to improve.

The deeplabv3 work has shown that the parallel structure is performed better than the sequential structure. The ASPP module is proposed. We also adopt a parallel manner to cope with input feature maps. However, to better extract the hierarchical contexts, we do not add or concatenate the feature maps from convolutions with different dilated rates.

Because the remote sensing image has the property of high-intraclass variance and low-class variance, the standard multiscale module is not powerful enough, which means that the very local information is not enough for classifying

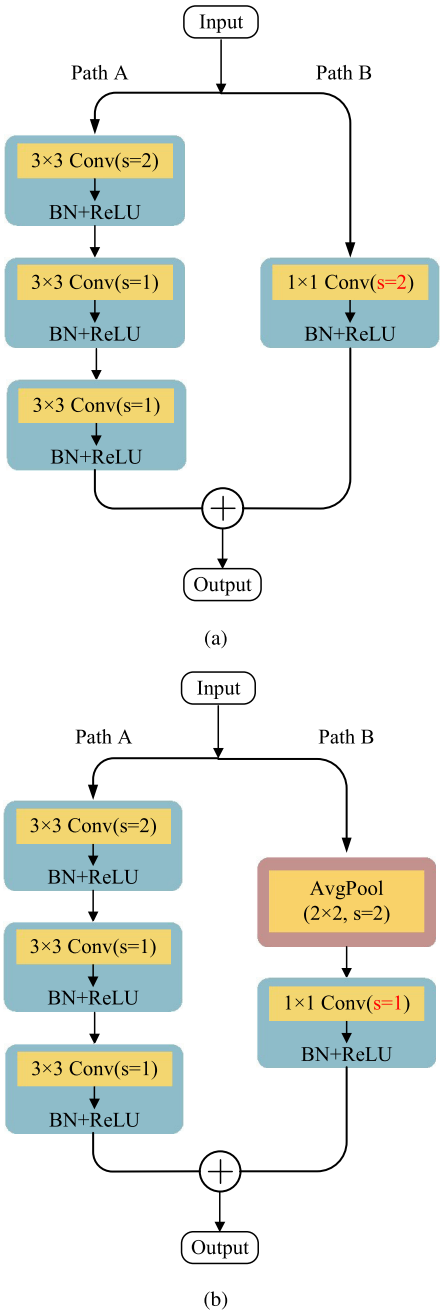


Fig. 4. ResNet tweaks. (a) Standard down-sampling unit in ResNet structure. (b) Proposed correction that adds an average pooling layer in front of the 1×1 convolution in the right path.

a target identification. Both the ASPP module and the SPP module do not notice their hierarchical dependencies in contexts of different scale output. To fix this tough problem, the larger contextual information on the feature maps should be used. In our work, a new multiscale module is proposed. The manner of aggregating hierarchical information is from global to local. The larger area contextual information is used to help improve the accuracy in a smaller context. Such a structure can make the most of the hierarchical dependencies of multiscale contexts. For example, the window could be on both cars and the roof. However, if a row of windows is detected, the object should be identified as a roof.

The hierarchical outputs from the ASPP layer are fused from global to local, and then, the output feature maps will be multiplied by the attention features, which is the encoder final features passing through a 1×1 convolution. Finally, the global average pooling branch will be added with the output feature maps, which can improve the accuracy further. This attention module can help recover the detailed information. Besides, the channelwise attention features are produced by the global pooling branch, which is fused with the spatialwise attention output. The computing process can be formulated as follows:

$$y(i) = C\{\cdots[\sum_{k}(x(i + r_1 \times k)) \cdot w(k) + \sum_{k} x(i + r_2 \times k) \cdot w(k)] + \cdots\} \cdot C[x(i)] + G[x(i)] \tag{2}$$

where r_i denotes different dilated rates, $x(i)$ denotes pixel in input feature maps, and $\sum_k(x(i + r_1 \times k)) \times w(k)$ denotes the output of dilated convolution. $C[\cdot]$ denotes the 1×1 convolution, and $G[\cdot]$ denotes the global average pooling. $y(i)$ denotes the output feature maps.

Fig. 5 shows the whole PAP module. Outputs of dilated convolutions will be added from high dilated rate to low. Besides, before the next addition, the 1×1 convolution is adapted to integrate the features.

To better extract multiscale context features, based on experimental experience, we use 3×3, 6×6, 12×12, and 18×18 dilated convolution layers in the PAP module. Despite the large kernel size, the dilated convolution does not introduce too many computations. Then, the parallel output results of the ASPP module are fused in order of largest to the smallest convolutional kernel. The fused result is multiplied by the encoder feature maps passing through a 1×1 convolution. Finally, the global average pooling could produce the weight vector with the global context, which could help select the essential features.

C. Attention Up-Sample Module

Low-level feature maps contain lots of detailed information, such as shape, boundary, and area, while high-level feature maps extract more semantic information, which can classify the pixel category. In the decoder phase, there are alternative up-sample methods, such as deconvolution [31], up-pooling, and interpolation algorithm. Empirically, bilinear up-sampling is effective and computing saving. Thus, in our work, the bilinear up-sampling layer helps recover the high-level feature maps to the original shape. However, due to the lack of detailed information from low-level feature maps, the model performance will be limited. As introduced in Section II, lots of models are proposed to fix the problem. Like U-Net, the skip connection module is adopted in this work. The low-level feature maps from the encoder will be concatenated with high-level feature maps. Most models use adding the operation to aggregate the information from low- and high-level features. However, like the standard ASPP module, the hierarchical dependencies among the low-level features and high-level features are ignored. The model performance is also limited.

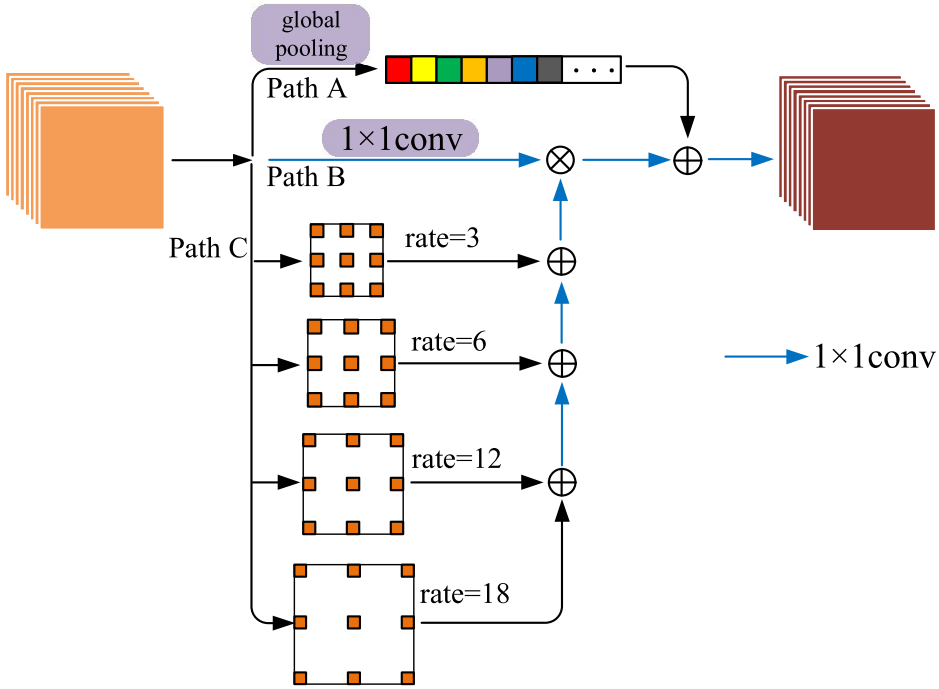


Fig. 5. PAP module. There are three paths in the PAP module. Path A adopts global average pooling adding with the output. Path B makes the origin feature maps from the encoder pass through a 1×1 convolution. Path C extracts multiscaled context information by different scale dilated convolutions and aggregates them from global to local.

To extract hierarchical information better, the channelwise attention mechanism is introduced.

Besides, it is difficult to decide the contributions of low-level features from the encoder and high-level features from the decoder. Thus, the channelwise attention-based method will help to learn an important weight. Passing through a global average pooling layer, the high-level feature maps can produce a weight vector, which can guide the low-level feature maps to select the most effective features, by multiply. In another word, there is plenty of semantic information in high-level feature maps, which could help label pixel categories, while there are lots of detailed information in the low-level feature maps, which could give the location and boundary information. The module is to recover the low-level information under the guidance of semantic information. The process can be formulated as follows:

$$y(i) = x_{\text{low}}(i) \cdot G[x_{\text{high}}(i)] + x_{\text{high}}(i) \tag{3}$$

where $x_{\text{low}}(i)$ denotes low-level features from encoder, $x_{\text{high}}(i)$ denotes high-level features from decoder, $G[\cdot]$ denotes the global average pooling, and $y(i)$ denotes output features.

The global context information could be offered by the global average pooling layer. Then, the low-level features could select the category localization details. Generated from the global average pooling layer, the weight vector will be pass through a 1×1 convolution and an ReLU activation. Finally, the weighted low-level features will be added with high-level features. Fig. 6 shows the process clearly.

D. PIC Module

Pooling layers in CNN models can help gain rotation invariance and aggregate the information in the local area, such as the max-pooling layer and average pooling layer. However, the pooling operations also bring the problem of accuracy loss in semantic segmentation tasks because a large number of pixels are discarded during the forward propagation process. This also makes it difficult to recover the detailed information in the decoder phase for a semantic segmentation task.

Various methods are studied to recover the lost information in the encoder phase. SegNet is the first network that adopts the encoder–decoder framework. What is more, SegNet records the pooling index in memory when doing the max-pooling process in the encoder phase. Then, in the decoder phase, the output feature maps will be resized to the original image size by means of up-pooling, which restores each pixel to the original position in bigger feature maps according to the pooling index, because the pooling index records the position information of the distinct feature, which could help restore the detailed information. However, the sequent complicated convolutions introduce lots of computations.

To recover the detailed information as much as possible and saving, the model computes at the same time. The pooling index should be used more efficiently. The PIC module is, therefore, proposed to restore the low-level information of the raw image. The proposed module is presented in Fig. 7, which can be formulated as

$$y_{n+1}(i) = R[x_{\text{low}}(i)] + B[x_n(i)] \tag{4}$$

where $x_{\text{low}}(i)$ denotes the pixel i in low-level feature map from encoder, $x_n(i)$ denotes the pixel i in high-level feature map from decoder, and $y_{n+1}(i)$ denotes the output feature map. $R[\cdot]$ denotes the up-pooling operation, and $B[\cdot]$ denotes the process of bilinear resize.

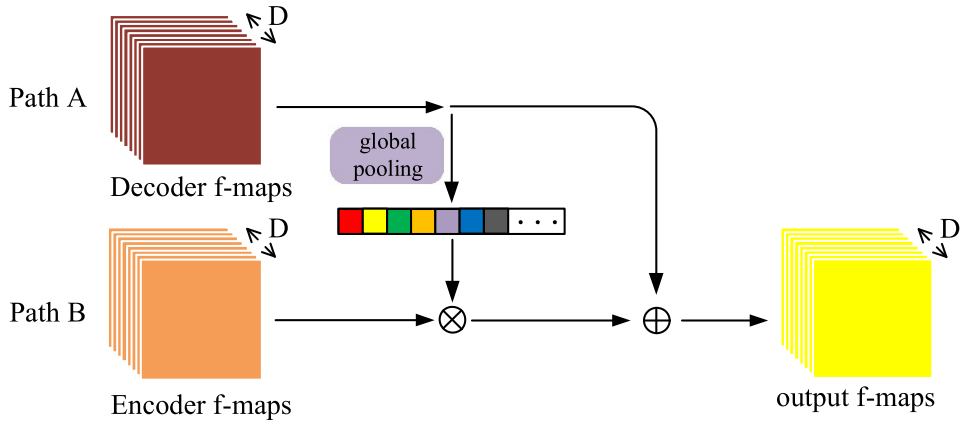


Fig. 6. AU module. There are two paths in the module. Path A inputs the high-level feature maps from the decoder, which produces a channel weight vector by passing through a global average pooling layer. Path B inputs the low-level feature maps from the encoder, which multiplies with weight and adds the path A input to obtain the output of this module.

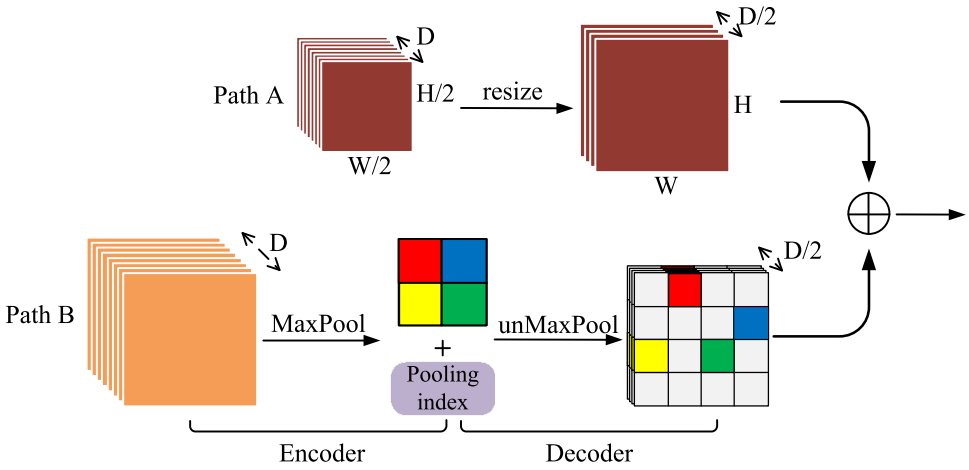


Fig. 7. PIC module. In this module, there are two paths. Path A inputs the high-level feature maps from the decoder, which will be recovered by the bilinear up-sampling. Path B inputs the low-level feature maps of the same size as path A from the encoder. Path B recovers the up-pooling map according to the pooling index.

In the proposed model, the backbone is ResNet-101 with dilated convolutions. There is only one max pooling layer in front of the first residual block, which makes the feature maps' size from 1/2 of the raw image to 1/4. Fig. 7 shows the process.

High-level feature maps from the decoder are resized by the bilinear up-sampling method. Low-level feature maps from the encoder are recovered by the up-pooling layer according to the pooling index in memory. Finally, two computing results are fused by the sum operation.

IV. Experimental Results

In this section, training details and the results will be given. The methods of data augmentation are introduced carefully here.

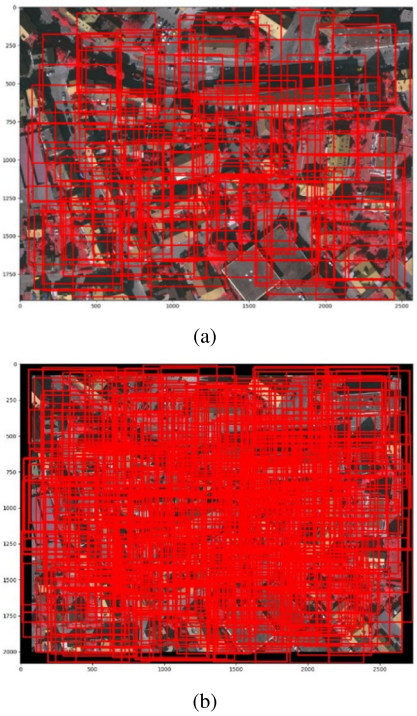
A. Dataset Details

The ISPRS Vaihingen dataset contains 33 orthophoto maps and related DSMs. Sixteen of them are labeled for training. The resolution is 9 cm, and the image size is over 2000×2000 pixels on average. Three bands are included in the image, including near-infrared, red, and green bands. Moreover, it includes six categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. To make a more generalized model for the semantic segmentation tasks of remote sensing images, the data of the DSM layer are ignored in our work.

B. Data Augmentation

A large amount of training data is essential when it comes to training a deep learning model. However, under the condition of limited training images, the number of training images must be increased with data augmentation skills. The size of a single image is almost 2000×2000, which is far beyond the GPU memory capacity. Thus, the patch-based training strategy is considered.

Most works crop the training image without overlap or with fixed overlap size, which makes the pixels in boundary share limit receptive field during all the training phase. To settle the



(a)

(b)

Fig. 8. Comparison between different sampling policies. (a) Random crop 100 patches from raw image. (b) Random crop 350 patches from images with 80 pixels padded.

problem, the random sampling with padding policy is proposed to generate the training dataset.

1) Random Sampling and Vertical/Horizontal Flip: Under the GPU memory limits, we crop the raw training image to 336 × 336 pixels. To alleviate the impact of fixed localization of a certain pixel, the random sampling method is adopted. The purposes are given as follows.

1) Make each pixel could be sampled.
2) Make each pixel randomly appear in a patch.

Hundred patches are cropped from one raw training image; the result is shown in Fig. 8(a). It is obvious that lots of pixels are ignored. Because lots of pixels at the boundaries are not noticed, as shown in Fig. 8(a), 80 pixels padding are added around the raw image to ensure that every pixel in the raw image could be sampled. First, we tried to crop 350 training patches from the raw image. However, Fig. 8(b) shows that too many bounding boxes are overlapped with each other. Finally, the sampling number is set to 200 as a tradeoff.

Then, for the same patch, the vertical flip and horizontal flip operations are carried. After all training images are sampled, we get 25 600 training patches for the ISPRS Vaihingen dataset. For the test image, the sliding window method is adopted with the 1024 window size and 50% overlap.

2) Random Rotation: When the training dataset is built, the vertical flip and horizontal flip operations have been processed to enhance the rotation invariance of the proposed model. However, only these four degrees (0, 90, 180, 270) are not enough to enhance the model's robustness to different angles. The patches are also randomly rotated from 0° to 45° for each training process.

3) Random Scale and Random: In the above operations, a random sampling method makes each pixel in the raw image appear in a different position of each patch randomly, while the random rotation method improves the ability of rotation invariance of the model to the input data. However, these operations are not sufficient to make a robust and accurate model. Besides, 512 × 512 size is still a computing burden for our training device. Thus, the random scale method and the random crop method are adopted before the training patches are fed into the model. The 512 × 512 patches will be scaled from 0.5 to 2 times randomly by bilinear interpolation. Then, it will be randomly cropped to 336 × 336 pixels to match the GPU memory.

Some works using the random resize method, which changes the aspect ratio of the image. However, this method destroys the geometric relationship in the remote sensing image, which makes the different distribution of input data between the training image and the testing image. For the method of random scale, it changes the receptive field of the model, even for the same training image, which helps the model extract information from multiscale contexts. After a series of data augmentation processes, the training image patches are shown in Fig. 9. It can be found obviously that the size of the car category in 9(b) and (c) is different. Every patch is rotated more or less.

C. Training Details

In our experiment, considering the model generalization, the DSM layer is not taken into account in our model. Because of the limitation of the number of training images, some data augmentation techniques are adopted to the original images. Training image patch size is 336 × 336 pixels. The SGD optimizer is adopted with the weight decay of 0.0005 for normalization. The bias in our model is not normalized. The multistage learning rate adjustment policy is adopted to drop the learning rate by 10 in every 15 epochs, from 0.01, with 60 epochs. The batch size is 6. All the experiments are processed in an NVIDIA GTX 1080Ti.

D. Results on the Vaihingen Dataset

Table I reports the semantic result on the ISPRS Vaihingen dataset. Like most models, the clutter/background class is ignored because of the very few pixels.

The evaluation for each class is based on the computation of F1-scores. The F1-score is defined as the harmonic mean of precision and recall

$$\text{precision} = \frac{\text{tp}}{\text{tp} + \text{fp}} \tag{5}$$

$$\text{recall} = \frac{\text{tp}}{\text{tp} + \text{fn}} \tag{6}$$

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{7}$$

where tp means the true positive, fp means the false positive, and fn means the true negative.

Mean-IOU and accuracy scores are calculated to measure the performance of the model for all categories.

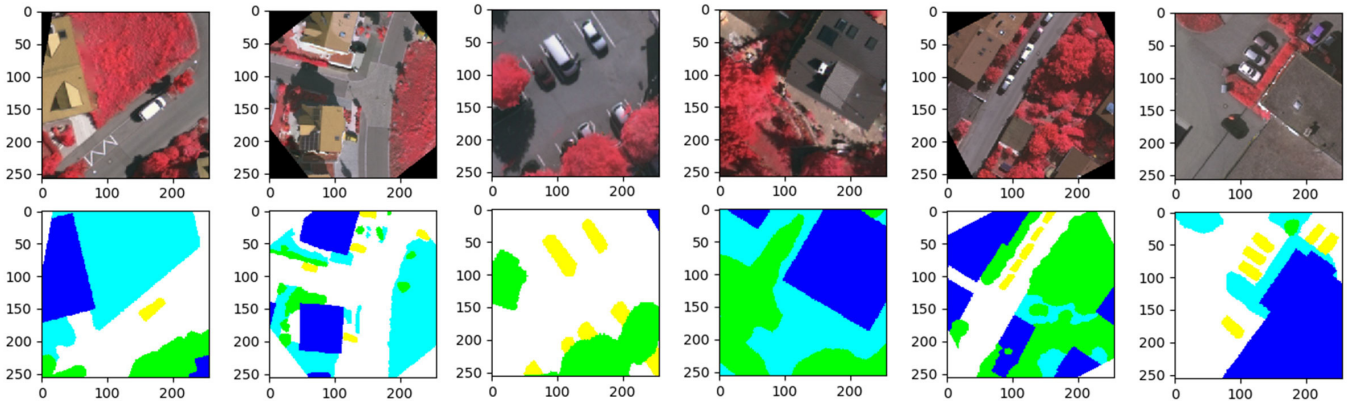


Fig. 9. Training patches after the data augmentation process.

TABLE I
RESULTS ON THE ISPRS VAIHINGEN DATASET

Model	Impervious surfaces	Building	Low vegetation	Tree	Car	Mean IOU	Overall acc
FCN-32s [3]	82.12	83.71	71.34	76.09	23.5	54.19	78.42
FCN-8s [3]	83.31	84.52	77.36	79.26	55.83	64.57	81.01
SegNet [5]	86.72	88.57	78.29	79.56	65.76	66.05	83.97
RefineNet [13]	87.22	87.15	79.41	74.86	75.45	68.11	84.79
PSPNet [23]	88.70	91.95	78.34	85.18	69.93	72.70	86.65
DeepLabv3+ [20]	89.20	93.10	**80.62**	85.66	74.05	75.12	87.52
SCAttNet [37]	89.13	90.30	80.04	80.31	70.50	70.2	85.46
ReSegNet [39]	**90.10**	92.60	78.70	85.50	75.92	74.63	87.16
SSAtNet(ours)	90.05	**93.66**	80.34	**86.48**	**81.20**	**76.40**	**88.01**

TABLE II
ABLATION EXPERIMENTS (%) ON THE ISPRS VAIHINGEN DATASET

Model	Impervious surfaces	Building	Low vegetation	Tree	Car	Mean IOU	Overall acc
Baseline	89.52	93.21	**80.40**	85.46	74.29	74.85	87.22
+PAP	89.80	93.20	80.13	85.18	77.11	75.46	88.02
+PAP+AU	**90.08**	93.50	80.25	86.04	79.72	76.01	87.97
+PAP+AU+PIC	90.05	**93.66**	80.34	**86.48**	**81.20**	**76.40**	**88.01**

Calculating accuracy is to evaluate whether each pixel on the predicted output and the ground truth is matched, while mean-IOU is to evaluate whether the proposed model could predict a more accurate shape of the object. Thus, the mean-IOU score could reflect the model performance batter.

In Table I, the performances of several perfect networks are listed. The overall accuracy gives the result of how many pixels matched, while the F1-score focuses more on the low-accuracy category, which can measure the model performance more precisely.

From the experimental result, we could find that the overall accuracy of SSAtNet is improved 0.56% compared with the DeepLabv3+ network, while the mean-IOU of SSAtNet is improved by 1.68%, which could present the model performance more accurately. Besides, ReSegNet is the best network for the dataset so far. However, under the condition of our training parameters, the performance is improved 0.98% in overall accuracy and 2.37% in mean-IOU compared with it.

Also, the performance on the car category of FCN-32s is quite low, which makes both the mean-IOU and overall accuracy a low accuracy. However, the accuracy of the car on FCN-8s improved greatly, which leads to a relatively high overall accuracy. We could find that the low-level features work. The proposed SSAtNet achieves quite high accuracy in the car category, improving 6.95% compared with the ReSegNet. It shows that SSAtNet could recognize small objects effectively. All results of different models are shown in Table I.

E. Ablation Study

To evaluate the performance brought by each proposed module, the ablation experiments of the model are carried. Table II lists the performance of the different modules. The DeepLabV3 [31] model is set to be the baseline model. The encoder is based on the ResNet-101 with dilated convolutions, while the multiscale module is a standard ASPP layer. The PAP module, the AU module, and the PIC module are introduced into the baseline model progressively.

Generally speaking, each of these modules improves the effectiveness of SSAtNet progressively. The proposed SSAtNet exceeds state of the art in many categories, such as building, tree, and car. Besides, we could find that all these modules have contributed to the improvement of the car

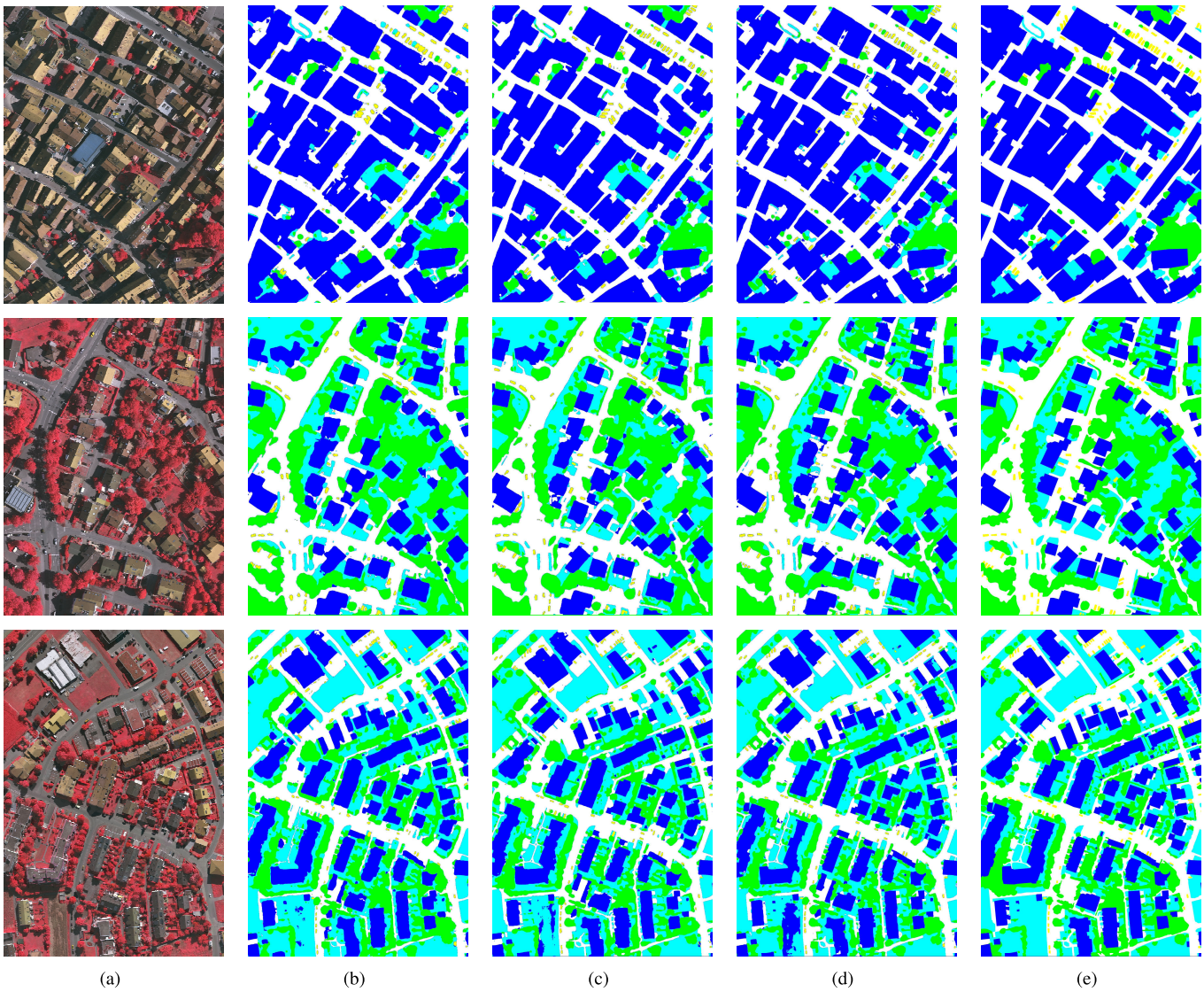


Fig. 10. Performance comparison between different models on the ISPRS Vaihingen dataset. (a) Raw image. (b) Predicted color map of model PSPNet. (c) Predicted color map of model ReSegNet. (d) Predicted color map of model SSAtNet. (e) Ground truth.

category, which is the most difficult category to be captured. It can be calculated that a 9.3% improvement in the accuracy of the CAR category is achieved.

F. Discussion

The proposed SSAtNet improves the semantic segmentation accuracy in both F1-score and overall accuracy. In this section, the predicted map is visualized for intuitive analysis. Our work selects the predicted map from three models, including PSPNet, ReSegNet, and the proposed SSAtNet. Three predicted maps of test images are selected to be presented in Fig. 10. Roughly speaking, the predicted maps of these three test images are quite similar. However, the proposed SSAtNet could predict a more accurate shape, which could be seen in the building category.

The same area in three different predicted maps is cropped for comparison, which is presented in Fig. 11. From the comparison, we could find that the edge of objects is located more accurately in the SSAtNet result. The outline of category building in Fig. 11(d) is clearer than both (b) and (c). The sides and corners of the building can be distinguished. This result shows that the proposed model can distinguish the contours of trees and buildings more effectively.

Besides, the judgment of the car category is more accurate. The false positives and true negatives samples in SSAtNet results are declined.

Both the data and the visual enhancements indicate that the proposed model can effectively capture multiscale contexts in a large scope. It also means that making full use of the multiscale contexts can improve the baseline model performance indeed. In the proposed model, attention mechanisms are introduced into the ASPP module, with a global-to-local fusing strategy, to capture the multiscale contexts. What is more, the AUs module and the PIC module can focus more on the low-level features, which can enhance high-level contextual information greatly.

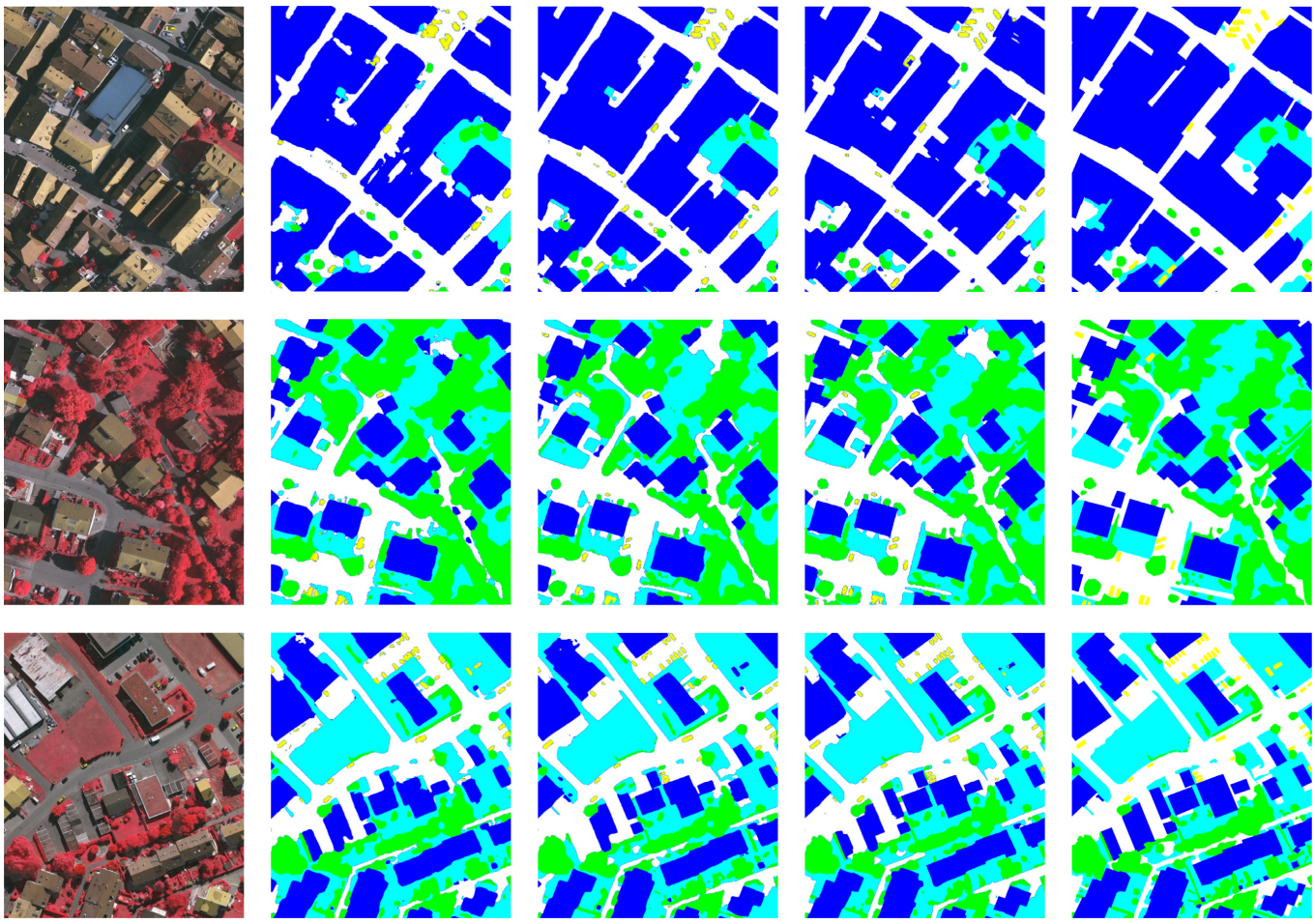


Fig. 11. Performance comparison of a patch between different models on the ISPRS Vaihingen dataset. (a) Raw image. (b) Predicted color map of model PSPNet. (c) Predicted color map of model ReSegNet. (d) Predicted color map of model SSAtNet. (e) Ground truth.

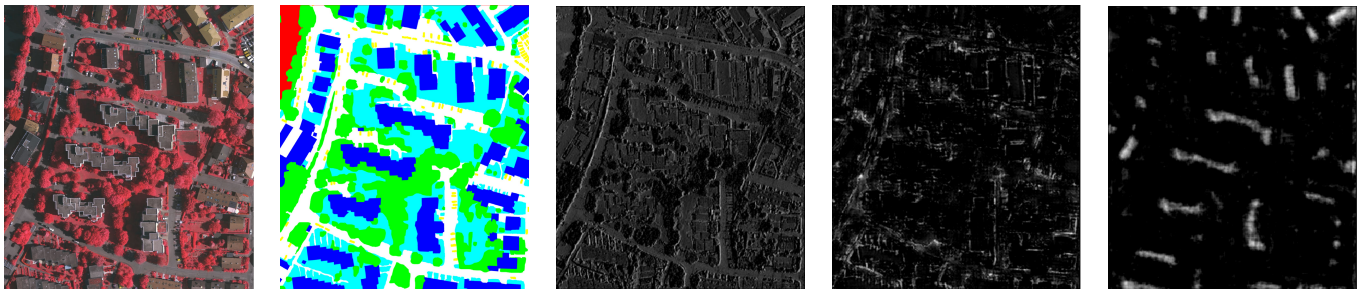


Fig. 12. Visualization of the feature maps in our model. (a) Raw image. (b) Ground truth. (c) Output feature map in the first block of the ResNet-101. (d) Output feature map of the encoder. (e) Feature map of the attention layer.

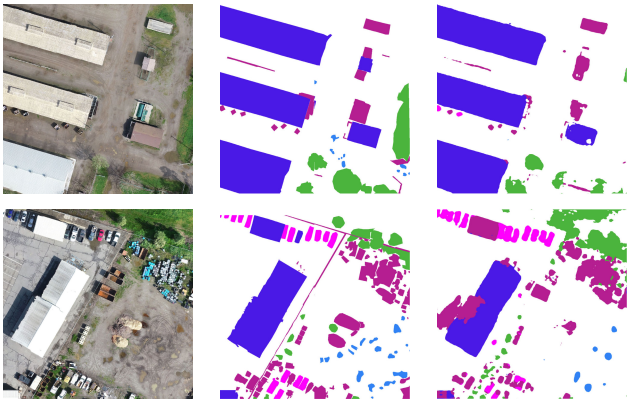
The most important module in our proposed model is PAP, which integrates the multiscale contexts to make the model could pay more attention to the different targets. To check the effect of the attentional mechanisms, which are used in our model, we visualize the feature maps of different layers, as shown in Fig. 12. The image is for test purpose. Fig. 12(b), which is the ground truth, shows different categories with different colors clearly. Fig. 12(c) and (d) shows feature maps from the ResNet-101 backbone, which shows the process of feature capture. Fig. 12(e) shows the feature map from the attention layer. As can be seen, this feature map is mainly used for the identification of the category of buildings. The lighter area indicates that the category of the pixel may be the building.

G. Results on the Dronedeploy Dataset

We conduct the complement experiment on the dronedeploy dataset. The explanation and results have been added in the following. The "dataset-medium" is adopted for testing. However, for the time limit, we crop the raw image as

TABLE III
RESULTS (IOU/F1-SCORE) ON THE COMPLIMENTARY DRONEDEPLOY DATASET

Model	BUILDING	VEGETATION	WATER	GROUND	CAR	Mean IOU	Overall acc
Baseline	45.86/66.73	52.77/70.08	35.52/43.9	82.12/88.95	48.04/66.90	47.03	84.12
+PAP	48.00/67.18	56.01/71.83	37.67/44.12	85.47/90.81	51.23/68.01	51.80	86.31
+PAP+AU	51.29/69.10	56.77/72.08	38.52/44.90	87.12/92.75	52.04/68.90	53.03	87.62
+PAP+AU+PIC (SSAtNet)	52.32/68.71	58.00/74.53	39.10/45.02	87.43/93.19	56.71/72.00	56.21	88.23



(a) (b) (c)

Fig. 13. Visualization of the feature maps in our model. (a) Raw image. (b) Ground truth. (c) Predicted color map.

the sample code method. In another word, the random crop method is not performed on this dataset. The scores of "cluster" and "ignore" categories are not calculated in the model performance. The set of our experiment is listed as follows.

1) Training and test images are cropped into 336∗336 size.
2) The SGD optimizer is used with a batch size of 6.
3) Multistage learning rate adjusting policy is adopted to decline the learning rate from 0.01 to 0.00001 with 20 total epochs.

The experimental results are presented in Table III, and since we did not find any other model performance on this dataset, only the performance of our proposed model is given. Fig. 13 shows the raw image, ground truth, and the predicted color map of our model.

Besides, we carry out the ablation study to evaluate the performance of the proposed module on the dronedeploy dataset. Also, the DeepLab V3 model is set to be the baseline model. It is obvious that the proposed modules could improve the model performance progressively by adding one by one. Especially, our model improves the accuracy of the CAR category over 18%. Compared to the baseline model, the performance of the SSAtNet on mean-IOU is improved over 19.5% and 4.8% on overall accuracy.

Considering the huge size of this dronedeploy dataset, we do not apply any data augmentation methods to the dataset. However, the performance that has been achieved illustrates that the proposed model can be applied for the task of semantic segmentation of remote sensing images in natural scenes.

V. CONCLUSION

In this work, a novel end-to-end attention-based semantic segmentation neural network is proposed to perform on remote sensing images for the image semantic segmentation tasks. For the character of large intraclass variance and interclass variance in remote sensing images, the spatialwise and channelwise attention mechanisms are introduced into the model to capture multiscale contexts. A PAP module is proposed to help the model refine object fine-grained information from global to local. In the decoder phase, the attention up-sampling module introduces channelwise attention to help the deeper feature maps recover detailed information better. Besides, to further alleviate the problem of pixel loss in the pooling layer, the PIC module is proposed to integrate high-level features with distinct low-level features. What is more, the ResNet-101 backbone is refined to capture the multiscale contexts and small features. Except for the model design, a series of data augmentation methods are adopted to make the model more robust. As a result of these designs, the experiments on the ISPRS Vaihingen dataset demonstrate that SSAtNet can capture multiscale information effectively, which outperforms the state-of-the-art methods. Especially for the car category, a great improvement is achieved. By our work, how to design a model on the basis of the current framework is still worth studying.

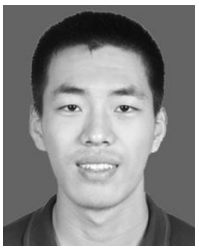
REFERENCES

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
[2] J. Zhao, X. Mao, and L. Chen, "Speech emotion recognition using deep 1D & 2D CNN LSTM networks," *Biomed. Signal Process. Control*, vol. 47, pp. 312–323, Jan. 2019.
[3] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[4] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.
[5] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
[6] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[7] A. Shrivastava, A. Gupta, and R. Girshick, "Training region-based object detectors with online hard example mining," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 761–769.
[8] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
[9] Z. Wu, C. Shen, and A. van den Hengel, "Wider or deeper: Revisiting the ResNet model for visual recognition," *Pattern Recognit.*, vol. 90, pp. 119–133, Jun. 2019.
[10] S. Zagoruyko and N. Komodakis, "Wide residual networks," 2016, *arXiv:1605.07146*. [Online]. Available: http://arxiv.org/abs/1605.07146
[11] M. A. Islam, M. Rochan, N. D. B. Bruce, and Y. Wang, "Gated feedback refinement network for dense image labeling," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3751–3759.

[12] C. Peng, X. Zhang, G. Yu, G. Luo, and J. Sun, "Large kernel matters—Improve semantic segmentation by global convolutional network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4353–4361.
[13] P. O. Pinheiro, T.-Y. Lin, R. Collobert, and P. Dollár, "Learning to refine object segments," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 75–91.
[14] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, "Gated-SCNN: Gated shape CNNs for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5229–5238.
[15] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*. [Online]. Available: http://arxiv.org/abs/1511.07122
[16] P. Krähenbühl and V. Koltun, "Efficient inference in fully connected CRFs with Gaussian edge potentials," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, pp. 109–117.
[17] S. Zheng *et al.*, "Conditional random fields as recurrent neural networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1529–1537.
[18] A. G. Schwing and R. Urtasun, "Fully connected deep structured networks," 2015, *arXiv:1503.02351*. [Online]. Available: http://arxiv.org/abs/1503.02351
[19] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.
[20] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
[21] S. Lazebnik, C. Schmid, and J. Ponce, "Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2, Jun. 2006, pp. 2169–2178.
[22] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[23] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[24] L. Chen *et al.*, "SCA-CNN: Spatial and channel-wise attention in convolutional networks for image captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5659–5667.
[25] F. Rottensteiner *et al.*, "The ISPRS benchmark on urban object classification and 3D building reconstruction," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 1, no. 1, pp. 293–298, 2012.
[26] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: http://arxiv.org/abs/1409.1556
[27] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 1–9.
[28] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[29] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.
[30] G. Huang, S. Liu, L. V. D. Maaten, and K. Q. Weinberger, "CondenseNet: An efficient DenseNet using learned group convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2752–2761.
[31] V. Dumoulin and F. Visin, "A guide to convolution arithmetic for deep learning," 2016, *arXiv:1603.07285*. [Online]. Available: http://arxiv.org/abs/1603.07285
[32] W. Liu, A. Rabinovich, and A. C. Berg, "ParseNet: Looking wider to see better," 2015, *arXiv:1506.04579*. [Online]. Available: http://arxiv.org/abs/1506.04579
[33] H. Li, P. Xiong, J. An, and L. Wang, "Pyramid attention network for semantic segmentation," 2018, *arXiv:1805.10180*. [Online]. Available: http://arxiv.org/abs/1805.10180
[34] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.
[35] G. Li, L. Li, H. Zhu, X. Liu, and L. Jiao, "Adaptive multiscale deep fusion residual network for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8506–8521, Nov. 2019.
[36] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2017.
[37] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, May 2021.
[38] X. Pan, L. Gao, B. Zhang, F. Yang, and W. Liao, "High-resolution aerial imagery semantic labeling with dense pyramid network," *Sensors*, vol. 18, no. 11, p. 3774, Nov. 2018.
[39] Y. Sun, Y. Tian, and Y. Xu, "Problems of encoder-decoder frameworks for high-resolution remote sensing image segmentation: Structural stereotype and insufficient learning," *Neurocomputing*, vol. 330, pp. 297–304, Feb. 2019.
[40] K. Nogueira, M. D. Mura, J. Chanussot, W. R. Schwartz, and J. A. dos Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, Oct. 2019.
[41] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.
[42] F. Zhou, R. Hang, and Q. Liu, "Class-guided feature decoupling network for airborne image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2245–2255, Mar. 2021.
[43] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

Qi Zhao (Member, IEEE) received the Ph.D. degree in communication and information systems from Beihang University, Beijing, China, in 2002.

She was a Visiting Scholar with the Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA, from 2014 to 2015. She is a Professor with Beihang University. Her research interests include image recognition and processing, communication signal processing, and target tracking.



Jiahui Liu received the B.S. degree in electronics and information engineering from Beihang University, Beijing, China, in 2018, where he is pursuing the M.S. degree with the School of Electronics and Information Engineering.

His research interests include deep learning, object detection, and remote sensing image segmentation.

Yuewen Li received the B.S. degree in electronics and information engineering from Beihang University, Beijing, China, in 2017, where he is pursuing the M.S. degree with the School of Electronics and Information Engineering.

His research interests include deep learning, object detection, and few-shot learning.

Hong Zhang received the B.S. degree from the Hebei University of Technology, Tianjin, China, in 1988, the M.S. degree from the Harbin University of Science and Technology, Harbin, China, in 1993, and the Ph.D. degree from the Beijing Institute of Technology, Beijing, China, in 2002, all in electrical engineering.

She was a Visiting Scholar with the Department of Neurosurgery, University of Pittsburgh, Pittsburgh, PA, USA, from 2007 to 2008. She is a Professor with the School of Astronautics, Beihang University, Beijing. Her research interests include activity recognition, image restoration, image indexing, object detection, and stereovision.